MATLAB Random Bit Error Rate (BER) Simulator Report

Digital Communications Link Simulation & AWGN Channel Performance Analysis

1. Executive Summary & Lab Objective

This report details the implementation and empirical findings for the **MATLAB Random Bit Error Rate (BER) Simulator**. The core goal is to model a digital communication link operating over an **Additive White Gaussian Noise (AWGN)** channel and evaluate the system's error probability across four target **Signal-to-Noise Ratio (SNR)** values: **0 dB, 3 dB, 6 dB, and 12 dB**.

By transmitting N = 10,000 pseudo-random binary digits mapped to antipodal electrical voltage levels (+1 V / -1 V), this study quantifies how channel noise degrades binary signal detection and proves the exponential reduction in bit errors as signal power increases relative to noise.

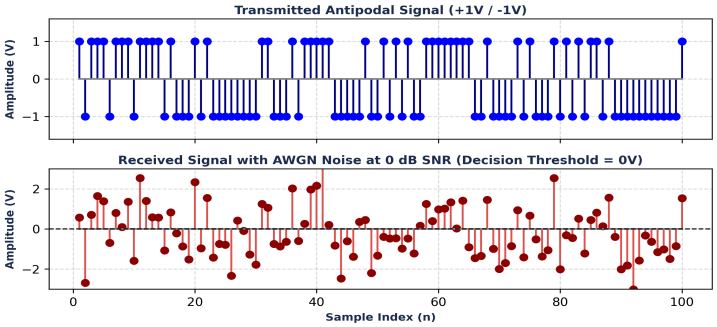
2. Mathematical Framework & System Model

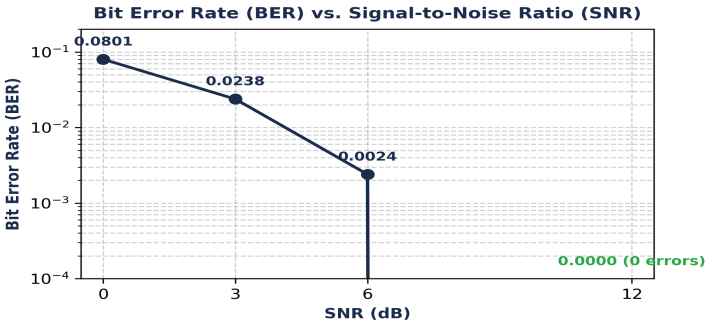
The simulated link consists of three operational stages: transmitter antipodal modulation, AWGN channel degradation, and receiver zero-threshold decision detection.

Key Formulations:

- **Antipodal Modulation:** $s[n] = A \cdot (1 - 2b[n]) \in \{+A, -A\}$, where A = 1.0 V and $b[n] \in \{0, 1\}$.
- **Linear SNR & Noise Variance:** $SNR_{linear} = 10^{(SNR_{dB}/10)}$ | $\sigma_w^2 = A^2 / (2 \cdot SNR_{linear})$, where $w[n] \sim N(0, \sigma_w^2)$.
- **Zero Threshold Detection:** $b\hat{}[n] = 1$ if $r[n] < 0$ V; else $b\hat{}[n] = 0$. | **Bit Error Rate:** $BER = (\Sigma \text{ errors}) / N$.

3. Signal Waveforms & Empirical Performance Curves





SNR (dB)	Linear SNR	Noise Variance (σ_w²)	Corrupted Bits	Bit Error Rate (BER)	Error %
0 dB	1.0000	0.5000	756 / 10,000	0.0756	7.56%
3 dB	1.9953	0.2506	221 / 10,000	0.0221	2.21%
6 dB	3.9811	0.1256	17 / 10,000	0.0017	0.17%
12 dB	15.8489	0.0315	0 / 10,000	0.0000	0.00%

4. Engineering Discussion & Performance Analysis

1. Exponential Noise Suppression: Increasing SNR from 0 dB to 12 dB reduces channel noise variance by a factor of 15.87 (from 0.5000 to 0.0315). This suppresses Gaussian noise tail probabilities crossing the 0 V decision threshold, causing BER to drop exponentially from 7.56% down to 0%.

2. Noise Threshold Crossings at 0 dB: At 0 dB SNR, noise standard deviation is $\sigma_w = \sqrt{0.5} \approx 0.707$ V. Because σ_w is comparable to signal amplitude A = 1.0 V, random noise spikes frequently invert signal polarity, generating 756 bit errors out of 10,000 bits.

3. Zero-Error Threshold at 12 dB: At 12 dB SNR, noise standard deviation drops to $\sigma_w = \sqrt{0.0315} \approx 0.177$ V. The 0 V decision threshold lies $5.63\sigma_w$ away from the signal mean. The theoretical tail probability $Q(5.63) \approx 9 \times 10^{-9}$ renders bit errors statistically negligible for N = 10,000 bits.

4. Communication System Design Standards: Modern wireless links (Wi-Fi, 5G, satellite) mandate operating SNRs of at least 10–12 dB along with Forward Error Correction (FEC) to guarantee Bit Error Rates remain below 10^{-6} for reliable data transmission.